



End Term (Even) Semester Examination May-June 2025

Roll no.

Name of the Program and semester: B.Tech (CSE) 4th Semester

Name of the Course: Finite Automata and Formal Languages

Course Code: TCS-402

Time: 3 hours

Maximum Marks: 100

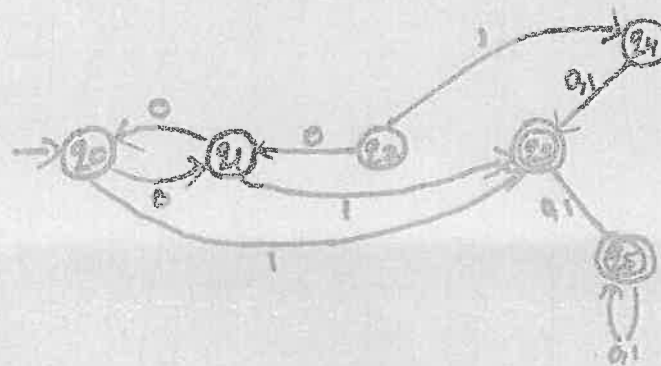
Note:

- (i) All the questions are compulsory.
- (ii) Answer any two sub questions from a, b and c in each main question.
- (iii) Total marks for each question is 20 (twenty).
- (iv) Each sub-question carries 10 marks.

Q1.

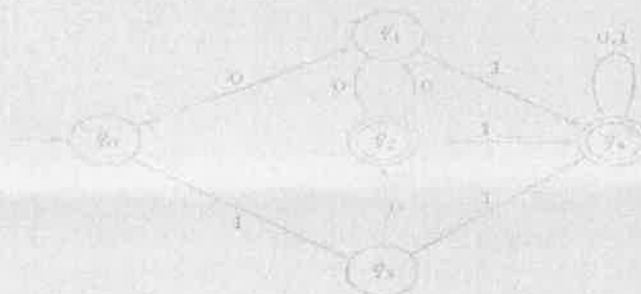
(2X10=20 Marks)

- a. Design a DFA that accepts all the strings over the alphabet $\Sigma = \{0, 1, 2\}$ such that the "number of 1's in the accepted string is a multiple of 4". [CO-2]
- b. Design a DFA that accepts all the strings over the alphabet $\Sigma = \{a, b\}$ such that the accepted string does not contains "aab" or "abb" as a substring. [CO-2]
- c. Apply Myhill-Nerode theorem to minimize the given DFA. [CO-2]

**Q2.**

(2X10=20 Marks)

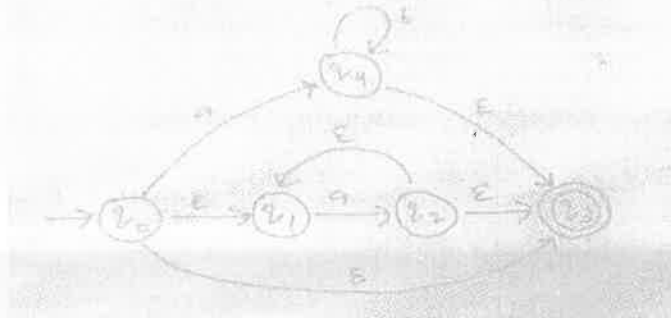
- a. Apply Arden's theorem to find the equivalent regular expression for the given finite automata. [CO-2]





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- b. Design a Mealy Machine that takes a binary number over the $\Sigma = \{0, 1\}$ as the input and finds the 2's complement as the output. Further, convert the resultant Mealy machine to Moore machine. [CO-6]
- c. Define ϵ -closure. Convert the given ϵ -NFA to equivalent DFA. [CO-2]



Q3:

- a. Convert the given CFG to CNF, where $V = \{S, A, B\}$ and $T = \{a, b\}$:

$S \rightarrow abAB$
 $A \rightarrow bAB \mid \epsilon$
 $B \rightarrow BAa \mid A \mid \epsilon$

(2X10=20 Marks)

[CO-3]

- b. Simplify the following CFG:

$S \rightarrow aA \mid aBB$
 $A \rightarrow aaA \mid \epsilon$
 $P \rightarrow bB \mid bbC$
 $C \rightarrow B$

[CO-3]

What language does the simplified grammar generates?

- c. Apply pumping lemma for CFL to prove that the given language L is not a context-free language:

$L = \{a^n b^n c^n \mid n = 2n\}$

[CO-3]

Q4:

- a. Convert the following CFG to PDA:

$S \rightarrow aAB \mid bBA$
 $A \rightarrow bS \mid a$
 $B \rightarrow aS \mid b$

(2X10=20 Marks)

[CO-4]

Also, test whether the string "abbaaabbbbab" would be accepted/rejected by the resultant PDA?

- b. Construct a PDA that recognizes strings from the given language:

$L = \{0^m 1^n 2^m \mid m \geq 1 \text{ and } n \geq 0\}$

[CO-4]

- c. Design a PDA that recognizes the following language by final state or by empty stack:

$L = \{w a^n w^R \mid w \in (a,b)^* \text{ and } n \geq 0, w^R \text{ is the reverse of } w\}$

[CO-4]



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Q5.

a. Write short notes on any two:-

(2X10=20 Marks)

- (i) Turing-Church Thesis
- (ii) Variants of Turing machine
- (iii) Post correspondence problem (PCP)

[CO-6]

b. Construct a Turing Machine for the given language over the $\Sigma = \{0, 1\}$:
 $L = \{WW \mid \text{where, } W \text{ is } (0 + 1)^*\}$

[CO-5]

c. Construct a Turing Machine that works as a copier. It takes a string W as the input and generates WcW as the output, where W is $(a, b)^*$

[CO-5]

Note For the question paper setters:

- Question paper should cover all the COs of the course.
- Please specify COs against each question.